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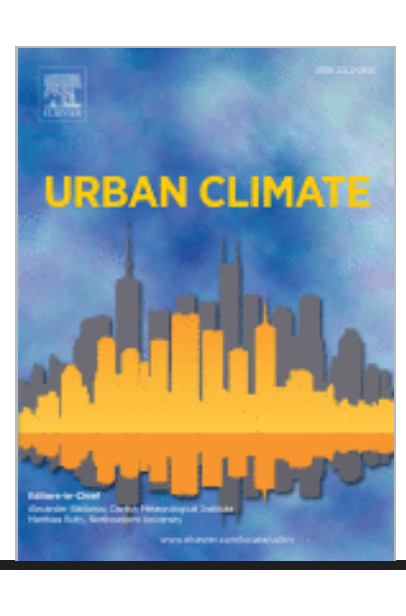

Urban Climate
Date: May 2023
Article: 101499
Volume: Volume 49
View accessibility information

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Machine learning applications on air temperature prediction in the urban canopy layer: A critical review of 2011–2022

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Abstract

Air temperature within the urban canopy layer is one of the most critical variables that impact the environmental sustainability of cities. With advantages in computational speed, machine learning approaches have been increasingly applied for urban air temperature prediction. Yet the current status and limitations of machine learning applications remain largely unclear. This study critically reviews 45 articles that use machine learning for urban air temperature prediction between 2011 and 2022. By separating the studies into two groups, we find that tree-based models fit for spatial prediction and neural network models work for temporal prediction. For spatial prediction, the root-mean-square error (RMSE) varies from 0.10 to 2.60 °C, and there is no clear relationship between RMSE and spatial resolution. Prediction errors of temporal studies tend to increase with the time horizon, with an RMSE of 1.35 ± 0.60 °C (4.02 ± 0.93 °C) for 4-h (5-days) ahead prediction. We provide three key recommendations to enhance the reliability and performance of future machine learning applications: 1) reporting metadata, quality control process, and model evaluation in detail, 2) fusing the temperature data from different sources and diversifying predictor variables, and 3) improving machine learning by stacking and integration with physical-based models.

Keywords

Urban heat island; Urban temperature; Machine learning; Artificial intelligence

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Introduction

The physical environment of global cities has changed significantly in response to rapid urbanization since the 1950s. With extensive evidence from the literature, the scientific community has reached a consensus that cities feature a distinct climate from their rural surroundings (Arnfield, 2003; Oke et al., 2017). The higher temperature in cities than in their rural counterparts, known as the urban heat island (UHI), has been reported over hundreds of cities worldwide (Stewart, 2011). Depending on the atmospheric layer, the UHI can be classified into boundary layer heat island (from the roof level to a height of 1–2 km) and canopy layer heat island (below the roof level). Air temperature in the urban canopy layer *(Ta)*, due to its important implications for urban sustainability and liveability, has been investigated the most among urban meteorological variables. On the one hand, *Ta* directly and indirectly influences human health. Studies have recognized clear associations between *Ta* and mortality (Anderson and Bell, 2009), and between *Ta* and urban pollutant concentration (Kalisa et al., 2018) as well as airborne diseases (Lu et al., 2021). On the other hand, *Ta* plays an essential role in determining the building energy consumption in cities (Li et al., 2019), which accounts for around 31% of global final energy use (Ürge-Vorsatz et al., 2012). It was reported that 1 °C rise of *Ta* could increase 0.5% to 8.5% of the electricity demand of buildings (Santamouris et al., 2015). Under climate change, the synergistic effect of local urban overheating and extreme heat waves will further exacerbate the thermal conditions in cities (Li and Bou-Zeid, 2013; Yang et al., 2019). Understanding the spatiotemporal pattern of *Ta* is critical, yet there is a lack of spatial *Ta* data in urban regions. Existing *Ta* measurements are typically from routine World Meteorological Organization certified stations operated by local weather service department. Though mobile transect measurements have been increasingly used to compensate for the limited spatial coverage by point-scale *Ta* data from weather stations (Heusinkveld et al., 2014; Yin et al., 2020; Kousis et al., 2021), it remains costly and challenging to obtain long-term high-resolution maps of *Ta* data across a city.

Timely and accurate prediction of *Ta* becomes essential for early warning, urban planning, and risk management services that support the development of climate-resilient cities. Predicting *Ta* is challenging due to the heterogeneous landscape, complex atmospheric dynamics, and various anthropogenic activities within individual cities. Large spatiotemporal variation in *Ta* has been observed in many cities (Cao et al., 2021; Nordbo et al., 2013; Tsin et al., 2016). For example, the difference in observed *Ta* among weather stations in Shenzhen, China at the same time (i.e., spatial variation) can reach >8 °C (Cao et al., 2021), while the diurnal variation of *Ta* can surpass 20 °C for weather stations in New York City (Ramamurthy et al., 2017). Empirical interpolation methods usually have limited capability to describe the spatial variability of *Ta* because ground-based stations are sparse in cities (Benali et al., 2012). Numerical weather prediction (NWP) using mesoscale models is the conventional approach for urban *Ta* prediction, and its accuracy has been steadily improved over the past decades (Bauer et al., 2015; Chen et al., 2011). However, several hurdles remain. First, to resolve the variation of *Ta* within cities, weather simulations need to incorporate boundary conditions and forcing data at a sub-kilometer spatial resolution, whose computational cost will impede the delivery of timely prediction. Second, existing parameterization schemes developed for physical processes at the mesoscale may not properly utilize the high-resolution input parameters. Recent studies suggested that refined input of three-dimensional urban canopy parameters would further amplify the positive bias in predicting urban wind speed (Li et al., 2021), which would affect the simulation of *Ta*. Third, predicted *Ta* from mesoscale weather models is unavoidably sensitive to small perturbations of the boundary conditions and forcing (Wu et al., 2005), where ensemble predictions are required to achieve a high level of confidence.

With the development of low-cost sensors, recent years have witnessed the establishment of dense urban climate monitoring networks in major metropolitan areas (Honjo et al., 2015; Lam et al., 2021; Tan et al., 2015). Observations from such networks can advance our understanding of *Ta* dynamics at the neighborhood scale, and the large databases enable the application of data-driven methods for *Ta* prediction. Unlike numerical weather simulations, data-driven models can provide fast predictions of *Ta* without in-depth knowledge of associated physical processes. When integrated with satellite-derived data and multi-platform observations, this approach is suitable for predicting *Ta* at the city scale (Alonso and Renard, 2019; Chen et al., 2022b). As a result, an increasing number of studies have examined the performance of machine learning (ML) algorithms in *Ta* prediction. However, the application area and study period vary vastly from study to study, and the understanding of ML model accuracy and limitation remains fractional. For example, for daily mean *Ta* prediction in the city of Turku in Finland, the boosted regression tree model performs well in July but badly in October (Hjort et al., 2011). Given the large difference in ML algorithms and site characteristics among existing studies, it is essential to examine the application quality of the literature and compare their findings for informing future model improvements.

To fill this gap, we conduct the first critical review of ML applications on *Ta* prediction, aiming to answer the following questions:

- What is the overall accuracy of machine learning methods for *Ta* prediction at a variety of spatiotemporal scales?
- Which machine learning model performs the best?
- What are the most important predictor variables?
- What is the limitation of current studies, and how to advance the *Ta* prediction in the future?

Section snippets

Literature sample inclusion criteria

The literature sample reviewed in this study was drawn from the English language, peer-reviewed articles published after year 2011. Web of Science was chosen as the search engine to retrieve relevant articles. We used three groups of keywords that correspond to machine learning (method), temperature (task), and urban (target area) during the search process. Several keywords were included in each group to retrieve a large number of related studies.

- Machine learning: machine learning OR artificial …

…

Overview of the literature sample

We first look into the study locations and periods of existing ML applications on urban air temperature. Fig. 2a shows the spatial distribution of study cities from the 45 reviewed articles. The majority of cities locate in East Asia, the Middle East, Europe, and North America. Seoul is the most studied city with six applications, followed by New York City which has four related studies. Africa, South America, and Oceania have few studies until now. It is noteworthy that three studies have …

Spatial prediction accuracy

The RMSE distribution of 28 ML applications on spatial prediction of *Ta* (from 28 out of 32 studies that clearly reported RMSE) at different spatial and temporal resolutions is summarized in Fig. 3. The RMSEs vary from 0.10 to 2.60 °C. The temporal resolution of existing studies ranges from the sub-hourly scale to the annual scale, with 27 applications concentrated on daily *Ta* (including daily mean, daily maximum, and daily minimum). In particular, daily maximum *Ta* has been investigated the …

Machine learning models

The machine learning models adopted in the reviewed 45 articles are summarized in Fig. 5. In total, there are 15 types of models, including Multiple Linear Regression (MLR), Support Vector Regression (SVR), Decision Tree (DT), Random Forest (RF), Extra Tree (ET), Gradient Boosting Decision Tree (GBDT), eXtreme Gradient Boosting (XGB), Stacking, Artificial Neural Network (ANN), simple Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Convolutional Neural …

Data

After summarizing the "state-of-the-art" models, the next question is how these models are driven. Generally speaking, the task of *Ta* prediction is to learn a mapping from the input data to output air temperature. The input data is called predictor variables or features, while the output *Ta* is known as the label data. The experience that a ML model learns is in the form of a set of N input-output pairs (N is the sample size, and is usually limited by available existing labels), with each pair …

Key recommendations

After reviewing the 45 literature samples, we have recognized several issues related to the reliability and comparability of machine learning applications on urban *Ta* prediction. Below we offer some key recommendations for improving the quality of ML applications, which aim to enhance the communication and understanding among researchers in the field of urban climate and atmospheric science. In this way, future applications can better leverage existing study findings and efficiently improve ML …

Concluding remarks

With the boost in data availability and computational resources in the past decade, there has been an increasing number of ML studies on urban climate. This study critically reviewed 45 articles over the past 12 years that applied ML models to predict *Ta* in urban areas. Based on the prediction task, studies were grouped into two general categories, spatial prediction and temporal prediction. The model algorithms, their corresponding accuracy and predictor variables were summarized, …

Nomenclature

ANN	Artificial Neural Network …
CNN	Convolutional Neural Network …
ET	Extra Tree …
DT	Decision Tree …
H_b	Building Height …
H_v	Vegetation Height …
FAI	Frontal Area Index …
F_b	Building Fraction …
F_r	Road Fraction …
F_v	Vegetation Fraction …
F_w	Water Fraction …
GAN	Generative Adversarial Network …
GBDT	Gradient Boosting Decision Tree …
GNN	Graph Neural Network …
GRU	Gated Recurrent Unit …
ISF	Impervious Surface Fraction …
NWP	Numerical Weather Prediction …
PDE	Partial Differential Equations …
Ta	Canopy Air Temperature …
T_B	Air Temperature Controlled by the Background …

…

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. …

Acknowledgement

This work was supported by the Hong Kong Research Grants Council funded project 16204220. …

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